AI & ML Final Project Report

Jazmín Calimano Colón

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PART I. Motivation

The Chicago Public Library are one of the few spaces in the city that provide free essential services to residents regardless of socioeconomic status. These spaces have evolved from being information repositories to centers of civic engagement, creativity, economic and workforce development, social cohesion, public health, and digital equity. It is crucial then for residents with the highest needs to maintain access to these services that otherwise would be unattainable due to a cost or knowledge barrier.

In this project I studied the factors that contribute to the amount of visits the Chicago Public Library branches receive and emerging patterns in the spatial distribution of these factors throughout the city's communities. Because Chicago's branch system was designed so that every community would have a library within walking distance, my aim was to inspect a dataset that had attributes on each branch and its surrounding neighborhood. My methodology for analysis included random forest and k-means clustering.

Essentially, my research objective was: *How does socioeconomic hardship shape library visitation across Chicago, and what geographic disparities emerge in that relationship?* I wanted to not just look at total foot traffic but understand why some branches were being used more than others and to identify where the greatest needs exist.

The open-source datasets that I used in this study included attributes monthly library visitation from the Chicago Data Portal, socioeconomic hardship indicators from the Great Cities Institute, and demographic measures from the American Community Survey. I focused on covering the years from 2013 to 2023 to have enough data points for the machine learning portion of my analysis.

PART II. Objectives

- Read, clean, and join into one dataframe the following datasets:
  - Chicago community boundary data
  - Branch location data
  - Chicago community hardship data
  - Library monthly visits data
  - Chicago community population data
- Complete feature engineering on final dataset
- Identify important factors with Random Forest
- Analyze spatial attributes with K-means Clustering
- Create Cluster profiles

PART III. Data

Packages and libraries

I will be using the following packages throughout this script:

```
library(pacman)
```

```
## Warning: package 'pacman' was built under R version 4.5.2
```

```
p_load(RSocrata, tidyverse, lubridate, sf, pdftools, tmap, stringr, ranger,
       ggplot2, dplyr, dendextend, skimr, ggrepel)
```

Spatial data

I want to look at how the libraries are distributed throughout the city of Chicago. For that, I'll be using the following data from the Chicago Data Portal (Library Locations). The data set contains a total of 81 rows and 10 columns. However, I am only interested in the branch name and the geographical location of the library.

According to this memo, several library branches experienced closures due to major renovations throughout the years (Memorandum). Additionally, libraries will occasionally experience emergency closures due to HVAC issues or utility interruptions.

```
socrata_file <- "https://data.cityofchicago.org/resource/x8fc-8rcq.csv"

lib_locations <- read.socrata(socrata_file)

head(lib_locations)
```

```
##          branch_
## 1    Albany Park
## 2        Altgeld
## 3 Archer Heights
## 4         Austin
## 5  Austin-Irving
## 6         Avalon
##                                                        service_hours
## 1 Mon. & Wed., 10-6; Tues. & Thurs., Noon-8; Fri. & Sat., 9-5; Sun., 1-5
## 2    Mon. & Wed., Noon-8; Tue. & Thu., 10-6; Fri. & Sat., 9-5; Sun., 1-5
## 3    Mon. & Wed., Noon-8; Tue. & Thu., 10-6; Fri. & Sat., 9-5; Sun., 1-5
## 4 Mon. & Wed., Noon-8; Tues. & Thurs., 10-6; Fri. & Sat., 9-5; Sun., 1-5
## 5    Mon. & Wed., Noon-8; Tue. & Thu., 10-6; Fri. & Sat., 9-5; Sun., 1-5
## 6    Mon. & Wed., Noon-8; Tue. & Thu., 10-6; Fri. & Sat., 9-5; Sun., 1-5
##                    address    city state   zip          phone
## 1       3401 W. Foster Ave. Chicago    IL 60625 (773) 539-5450
## 2          955 E. 131st St. Chicago    IL 60827 (312) 747-3270
## 3       5055 S. Archer Ave. Chicago    IL 60632 (312) 747-9241
## 4         5615 W. Race Ave. Chicago    IL 60644 (312) 746-5038
## 5   6100 W. Irving Park Rd. Chicago    IL 60634 (312) 744-6222
## 6 8148 S. Stony Island Ave. Chicago    IL 60617 (312) 747-5234
```

```
##                                website                branch_email
## 1 https://www.chipublib.org/locations/3/    albanypark@chipublib.org
## 2 https://www.chipublib.org/locations/4/       altgeld@chipublib.org
## 3 https://www.chipublib.org/locations/5/ archerheights@chipublib.org
## 4 https://www.chipublib.org/locations/6/        austin@chipublib.org
## 5 https://www.chipublib.org/locations/7/ austin-irving@chipublib.org
## 6 https://www.chipublib.org/locations/8/        avalon@chipublib.org
##                                          location
## 1 \n,  \n(41.97557881655979, -87.71361314512697)
## 2 \n,  \n(41.65719847558056, -87.59883829075173)
## 3 \n,  \n(41.80110836194246, -87.72648385568911)
## 4  \n,  \n(41.88917312206642, -87.7658203582574)
## 5 \n,  \n(41.95312267684315, -87.77928489355646)
## 6 \n,  \n(41.746393038286826, -87.5860053710736)
```

I created a smaller dataframe by selecting the necessary variables. We also need to check if th: `branch` and `location`, and I verified if there were any missing values.

```
locations_final <- lib_locations %>%
  select(branch = branch_,
         location = location)

any(is.na(locations_final))
```

```
## [1] FALSE
```

I transformed the location column values, which are currently a character class, into two separate numeric columns corresponding to latitude and longitude. Note that longitude will be the x-coordinate, and latitude will be the y-coordinate.

```
locations_final <- locations_final %>%
  mutate(location = str_trim(location)) %>%
  mutate(location = str_remove(location, "^,")) %>%
  mutate(location = str_trim(location)) %>%
  mutate(location = str_remove_all(location, "[()]")) %>%
separate(location, into = c("latitude", "longitude"), sep = ", ") %>%
  mutate(
    latitude = as.numeric(latitude),
    longitude = as.numeric(longitude)
  )
```

Then I turned the dataframe into a simple features spatial object using the coordinate reference system code of EPSG of 4326.

```
library.points <- st_as_sf(locations_final, coords = c("longitude", "latitude"),
                           crs = 4326, agr = "constant")
class(library.points)
```

```
## [1] "sf"         "data.frame"
```

Now that I have the exact point locations of the libraries, I need to layer a map of boundaries for each community the libraries belong to. I will download a map of community boundaries from the Chicago Data Portal using the sf read file function. Instead of coordinate points, the geometrical figures of the rows are polygons, which will delineate borders around each community.

```
comm.file <- "https://data.cityofchicago.org/resource/igwz-8jzy.geojson"
chicago.comm <- read_sf(comm.file)

chicago.comm <- chicago.comm %>%
  mutate(community = tolower(community))
```

The library points and the community area boundaries need to have the same projection, so they will be assigned the UTM (Universal Tranverse Mercator) zone 16N, which is the proper one for Chicago, with an EPSG code of 32616.

```
chicago.comm <- st_transform(chicago.comm, 32616)
library.points <- st_transform(library.points, 32616)
```

Chicago Community Hardship Data

The Great Cities Institute at UIC calculated the hardship index for each Chicago community using the ACS 5-year estimate surveys. Unfortunately, I couldn't find a machine readable dataset for the 2013 to 2017 calculation so I had to extract a PDF file for the information as follows. After reading the file, I parsed it by first turning each element into a separate list and then extracted the necessary parts.

```
pdf.file <- "2013to2017hardship_index.pdf"
hard.dat <- pdf_text(pdf.file)

nnlist <- ""
for (i in 1:3) {
  ppage <- strsplit(hard.dat[[i]],split="\n")
  nni <- ppage[[1]]
  nnu <- unlist(nni)
  nnlist <- c(nnlist,nnu)
}

nnlist <- nnlist[-c(1:5, 32:40, 67:74, 100:104)]
nnlist <- as.data.frame(nnlist)

nnlist <- nnlist %>%
  separate_wider_regex(nnlist, patterns = c(
      community           = "^[^0-9]+",
      area                = "\\s+\\d+",
      hardship_index      = "\\s+[0-9.]+",
      crowded_housing     = "\\s+[0-9.]+",
      hh_below_pov        = "\\s+[0-9.]+",
      unemployment        = "\\s+[0-9.]+",
      no_hs_diploma       = "\\s+[0-9.]+",
      pop_under18_over64  = "\\s+[0-9.]+",
      percapita_income    = "\\s+\\$[0-9,]+"))
```

Now before joining this hardship index dataframe to the 2019-2023 dataframe, each observational row needs to be identified per year.

```
years <- 2013:2017
```

```
nnlist_expanded <- nnlist %>%
  crossing(year = years)

index19_23 <- read.csv("Chicago_Hardship Index_2019-23.csv")

years2 <- 2019:2023

index19_23 <- index19_23 %>%
  rename(community = Community,
         area = Area.Number,
         hardship_index = Hardship.Index.Value,
         crowded_housing = Crowded.Housing.Rate..More.than.one.occupant.per.room.,
         hh_below_pov =
           Poverty.Rate..Share.of.households.with.income.below.poverty.level.,
         unemployment = Unemployment.Rate,
         no_hs_diploma = No.High.School.Diploma.Rate..Share.of.population.age.25.and.older.without.a.hig
         percapita_income = Per.Capita.Income) %>%
  crossing(year = years2)
```

Before joining the datasets together, all column class types need to be the same. After joining, all nonnumerical symbols should also be removed, such as the dollar sign in the income variable, then converted to a double type.

```
glimpse(nnlist_expanded)
```

```
## Rows: 385
## Columns: 10
## $ community          <chr> "Albany Park      ", "Albany Park      ", "Albany P~
## $ area               <chr> " 14", " 14", " 14", " 14", " 14", " 57", " 57", " ~
## $ hardship_index     <chr> "         45.7", "         45.7", "         45.7", ~
## $ crowded_housing    <chr> "        8.6", "        8.6", "        8.6", "     ~
## $ hh_below_pov       <chr> "        16.5", "        16.5", "        16.5", "  ~
## $ unemployment       <chr> "               7.1", "               7.1", "      ~
## $ no_hs_diploma      <chr> "             14.3", "             14.3", "        ~
## $ pop_under18_over64 <chr> "               32.0", "               32.0", "  ~
## $ percapita_income   <chr> "              $25,848", "              $25,848", "~
## $ year               <int> 2013, 2014, 2015, 2016, 2017, 2013, 2014, 2015, 201~
```

```
glimpse(index19_23)
```

```
## Rows: 390
## Columns: 10
## $ area               <chr> "1", "1", "1", "1", "1", "10", "10", "10", "10", "1~
## $ community          <chr> "Rogers Park", "Rogers Park", "Rogers Park", "Roger~
## $ crowded_housing    <dbl> 5.2, 5.2, 5.2, 5.2, 5.2, 1.1, 1.1, 1.1, 1.1, 1.1, 1~
## $ hh_below_pov       <dbl> 16.9, 16.9, 16.9, 16.9, 16.9, 7.7, 7.7, 7.7, 7.7, 7~
## $ unemployment       <dbl> 6.2, 6.2, 6.2, 6.2, 6.2, 4.5, 4.5, 4.5, 4.5, 4.5, 5~
## $ no_hs_diploma      <dbl> 10.3, 10.3, 10.3, 10.3, 10.3, 7.0, 7.0, 7.0, 7.0, 7~
## $ pop_under18_over64 <dbl> 26.4, 26.4, 26.4, 26.4, 26.4, 44.2, 44.2, 44.2, 44.~
## $ percapita_income   <chr> "$37,895", "$37,895", "$37,895", "$37,895", "$37,89~
## $ hardship_index     <dbl> 36.5, 36.5, 36.5, 36.5, 36.5, 30.1, 30.1, 30.1, 30.~
## $ year               <int> 2019, 2020, 2021, 2022, 2023, 2019, 2020, 2021, 202~
```

```
nnlist_expanded <- nnlist_expanded %>%
  mutate(area = as.character(area),
         crowded_housing = as.double(crowded_housing),
         hh_below_pov = as.double(hh_below_pov),
         unemployment = as.double(unemployment),
         no_hs_diploma = as.double(no_hs_diploma),
         pop_under18_over64 = as.double(pop_under18_over64),
         hardship_index = as.double(hardship_index),
         year = as.numeric(year))

index19_23 <- index19_23 %>%
  mutate(year = as.numeric(year),
         area = as.character(area))

hardship_index <- rbind(index19_23, nnlist_expanded)
```

As we can see from the list of frequencies for each community name and area, there are duplicates of values. Each library should have a total of 10 observations each corresponding to each year. Then the dataset should be inspected for missing values.

```
hardship_index$community <- trimws(hardship_index$community)
hardship_index$area <- trimws(hardship_index$area)

colSums(is.na(hardship_index))
```

```
##               area          community    crowded_housing       hh_below_pov
##                  0                  0                  5                  5
##       unemployment      no_hs_diploma pop_under18_over64   percapita_income
##                  5                  5                  5                  0
##     hardship_index               year
##                  5                  0
```

```
hardship_index <- na.omit(hardship_index)
```

Finally, the income per capita value needs to be converted into a numerical value, and the dollar sign and comma should be removed as follows.

```
hardship_index <- hardship_index %>%
 mutate(percapita_income = gsub("\\$", "", percapita_income),
        percapita_income = gsub(",", "", percapita_income),
        percapita_income = trimws(percapita_income),
        percapita_income = as.numeric(percapita_income))
```

Monthly Library Traffic Data

Every year, the Chicago Public Library publishes visitor traffic data for each branch and it is made available on the Chicago Data Portal. Since I wanted to analyze data from 10 different years, to simplify the reading and cleaning steps, I downloaded the files in bulk.

```
paths <- list.files("C:/Users/jazmi/OneDrive/Desktop/PA470 AI & Machine Learning/Research Project",
    pattern = "^Visitors_[0-9]{4}\\.csv$", full.names = TRUE)

visits <- paths %>%
  set_names(basename) %>%
  map(read.csv) %>%
  map(~ mutate(.x, across(everything(), as.character))) %>%
  list_rbind(names_to = "year") %>%
  mutate(year = parse_number(year)) %>%
  mutate(LOCATION = gsub("\\*$", "", LOCATION),
         YTD = gsub(",", "", YTD)) %>%
  mutate(across(everything(), ~ gsub(",", "", as.character(.x))))
```

Looking through the joined dataset of all the library visits throughout the years, I noticed that the data has not been consistently recorded in the same columns. For instance, there are two variables representing the zip code, ("ZIP" and "ZIP.CODE"), where the values are recorded alternatingly every year. This means I'll have to create new ones ensuring that the right values carry over.

```
visits <- visits %>%
  mutate(zipcode = coalesce(ZIP, ZIP.CODE)) %>%
  select(-c(ZIP, ZIP.CODE)) %>%
  mutate(geometry = case_when(
      str_starts(LOCATION, "POINT") ~ LOCATION,
      str_starts(Location, "POINT") ~ Location,
      TRUE ~ NA_character_)) %>%
  select(-Location) %>%
  mutate(across(where(is.character), str_trim))

visits <- visits %>%
  mutate(branch = case_when(
    !str_starts(LOCATION, "POINT") ~ LOCATION,
    !is.na(BRANCH) ~ BRANCH,
     TRUE ~ NA_character_)) %>%
  select(-c(LOCATION, BRANCH)) %>%
  mutate(branch = gsub("\\*$", "", branch))

visits <- visits %>%
  mutate(branch = case_when(
      branch == "Harold Washtington Library Center" ~ "Harold Washington Library Center",
      branch == "Woodson Regional Library" ~ "Woodson Regional",
      branch == "Daley Richard M.-W Humboldt" ~ "Daley Richard M.",
      branch == "Daley Richard J.-Bridgeport" ~ "Bridgeport",
      branch == "Daley Richard J." ~ "Bridgeport",
      branch == "Sulzer Regional Library" ~ "Sulzer Regional",
      branch == "Legler" ~ "Legler Regional",
      branch == "Roosevelt" ~ "Little Italy",
      TRUE ~ branch)) %>%
  mutate(across(everything(), tolower)) %>%
  select(-c(ADDRESS, CITY))
```

To prepare the dataset for machine learning analysis, I had to pivot longer so that each observation corresponded to the amount of visits each library branch received per month.

```
visits_long <- visits %>%
  pivot_longer(cols = 2:13,
               names_to = "month",
               values_to = "monthly_visits") %>%
  mutate(month = tolower(month))
```

Demographic Data

Finally, the last dataset I needed to download contained essential demographic information on each Chicago community.

```
area.pop <- read.csv("ACS_5_Year_Data_by_Community_Area.csv")
area.pop <- area.pop %>%
  mutate(Community.Area = tolower(Community.Area)) %>%
  rename(community = Community.Area)
```

Aggregating at the community level

To begin joining all information together, I first joined the chicago.comm dataset with the hardship_index dataset and the demographic data by their common variable "community". The values need to match exactly so I will change the letter cases for both.

```
hardship_index <- hardship_index %>%
   mutate(across(where(is.character), str_trim)) %>%
  mutate(community = tolower(community),
         community = case_when(
           community == "clearning" ~ "clearing",
           community == "lakeview" ~ "lake view",
           community == "o'hare" ~ "ohare",
           community == "o'hare" ~ "ohare",
           TRUE ~ community))

community_attributes <- hardship_index %>%
  left_join(area.pop, by = "community") %>%
  select(-ACS.Year)

community <- community_attributes %>%
  left_join(st_drop_geometry(chicago.comm), by = "community") %>%
  left_join(chicago.comm %>% select(community, geometry),
    by = "community") %>%
  st_as_sf()
```

Aggregating at the branch level

```
library.points <- library.points %>%
    mutate(across(where(is.character), str_trim)) %>%
  mutate(branch = tolower(branch),
         branch = case_when(
           branch == "harold washtington library center" ~ "harold washington library center",
```

```
            branch == "sulzer regional library" ~ "sulzer regional",
            branch == "daley, richard j." ~ "bridgeport",
            branch == "roosevelt" ~ "little italy",
            branch == "woodson regional library" ~ "woodson regional",
            TRUE ~ branch))

libraries <- visits_long %>%
  left_join(st_drop_geometry(library.points), by = "branch") %>%
  left_join(library.points %>% select(branch, geometry),
    by = "branch") %>%
  st_as_sf() %>%
   select(-geometry.x)
```

Combining the final dataset

```
unique_chicago_map <- community %>%
  select(community, geometry) %>%
  distinct(community, .keep_all = TRUE) # Keeps only 77 rows

fulldf <- st_join(
  libraries,
  unique_chicago_map,
  join = st_within
)

final <- fulldf %>%
  st_drop_geometry() %>%
  mutate(year = as.numeric(year)) %>%
  left_join(
    community %>% st_drop_geometry(),
    by = c("community", "year")
  )
```

Feature Engineering: Random Forest

I don't necessarily need all the variables in the full dataset because some are redundant and they might create bias in the random forest model. I ended up variables for analysis: - percent of white population - percent of black population - percent of asian population - percent of hispanic population - percent of youth population ($<$ 18 years old) - percent of senior population ($>$ 65 years old) - percent of working age population ($>$ 18 and $<$ 65) - percent of low income population ($<$ 50k) - percent of high income population ($>$ 50k) - population density These values represent a varied range of population characteristics related to race ethnicity, dependency status, and economic class.

```
final_engineered <- final %>%
  mutate(across(
    c(Total.Population, White, Black.or.African.American, Asian, Hispanic.or.Latino,
      Male.0.to.17, Female.0.to.17, Male.65., Female.65.., shape_area,
      Under..25.000, X.25.000.to..49.999, X.75.000.to..125.000, X.125.000..),
    readr::parse_number)) %>%
  mutate(
```

```
    Pct_White = White / Total.Population,
    Pct_Black = Black.or.African.American / Total.Population,
    Pct_Asian = Asian / Total.Population,
    Pct_Hispanic = Hispanic.or.Latino / Total.Population,
    Pct_Youth_0_17 = (Male.0.to.17 + Female.0.to.17) / Total.Population,
    Pct_Senior_65 = (Male.65. + Female.65..) / Total.Population,
    Pct_Working_Age = 1 - (Pct_Youth_0_17 + Pct_Senior_65),
    Pct_Low_Income = (Under..25.000 + X.25.000.to..49.999) / Total.Population,
    Pct_High_Income = (X.75.000.to..125.000 + X.125.000..) / Total.Population,
    Pop_Density = Total.Population / shape_area) %>%
  select(-c(YTD, crowded_housing, hh_below_pov, unemployment, no_hs_diploma,
            Record.ID, area_num_1, area_numbe, shape_area, shape_len, White, Black.or.African.American,
    American.Indian.or.Alaska.Native, Native.Hawaiian.or.Pacific.Islander,
    Other.Race, Multiracial, White.Not.Hispanic.or.Latino, starts_with("Male"),
    starts_with("Female"), Under..25.000, Total.Population, pop_under18_over64,
    X.25.000.to..49.999, X.50.000.to..74.999, X.75.000.to..125.000, X.125.000..))

final_engineered <- final_engineered %>%
  mutate(month = as.factor(month),
    area = as.factor(area),
    branch = as.factor(branch)) %>%
  mutate(monthly_visits = readr::parse_number(as.character(monthly_visits))) %>%
  filter(!is.na(monthly_visits)) %>%
  filter(branch != "harold washington library center")
```

Feature Engineering: Clustering

```
df_branch <- final %>%
  st_drop_geometry() %>%
  mutate(across(where(is.character) & !branch, ~ as.numeric(gsub(",", "", .x)))) %>%
  mutate(
    Pop_Density = Total.Population / shape_area,
    Pct_Youth = (Male.0.to.17 + Female.0.to.17) / Total.Population,
    Pct_Senior = (Male.65. + Female.65..) / Total.Population,
    Pct_Low_Income = (Under..25.000 + X.25.000.to..49.999) / Total.Population) %>%
  select(branch, YTD, hardship_index, Pop_Density, percapita_income, Pct_Youth,
         Pct_Senior, Pct_Low_Income) %>%
  filter(!branch %in% c("harold washington library center", "sulzer regional")) %>%
  group_by(branch) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE)) %>%
  ungroup() %>%
  filter(if_all(where(is.numeric), ~ !is.infinite(.))) %>%
  drop_na()

df_scaled <- df_branch %>%
  mutate(across(-branch, scale))
```

PART IV. Method

Random Forest

To predict library visits based on community metrics, I used the random forest method. To see which features actually mattered, I used permutation importance, which gave me a ranking of each predictor's impact. The plotted feature importance is in the following section.

```
set.seed(42)
rf_model <- ranger(
  formula = monthly_visits ~ .,
  data = final_engineered,
  num.trees = 500,
  mtry = 4,
  importance = 'permutation' )
```

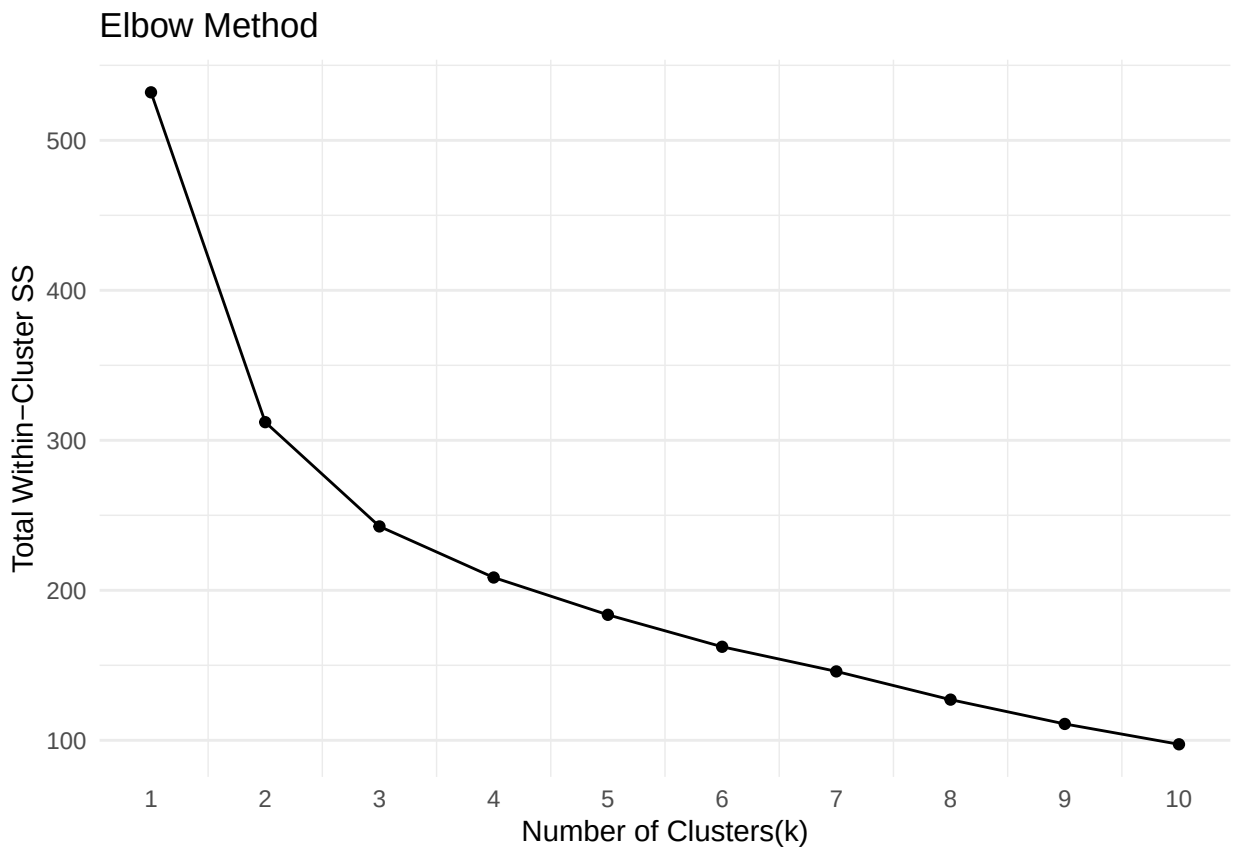
K-means Clustering

To determine the ideal number of clusters for the k-means, I need to calculate the total within-cluster sum of squares through the elbow method.

```
wss <- map_dbl(1:10, function(k) {
  kmeans(df_scaled %>% select(where(is.numeric)),
  centers = k, nstart = 25)$tot.withinss
})

elbow <- tibble(k = 1:10, wss = wss) %>%
  ggplot(aes(x = k, y =wss)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:10) +
  theme_minimal() +
  labs(title = "Elbow Method", x = "Number of Clusters(k)",
       y = "Total Within-Cluster SS") +
  theme(
    panel.grid.major.x = element_blank()
  )

elbow
```



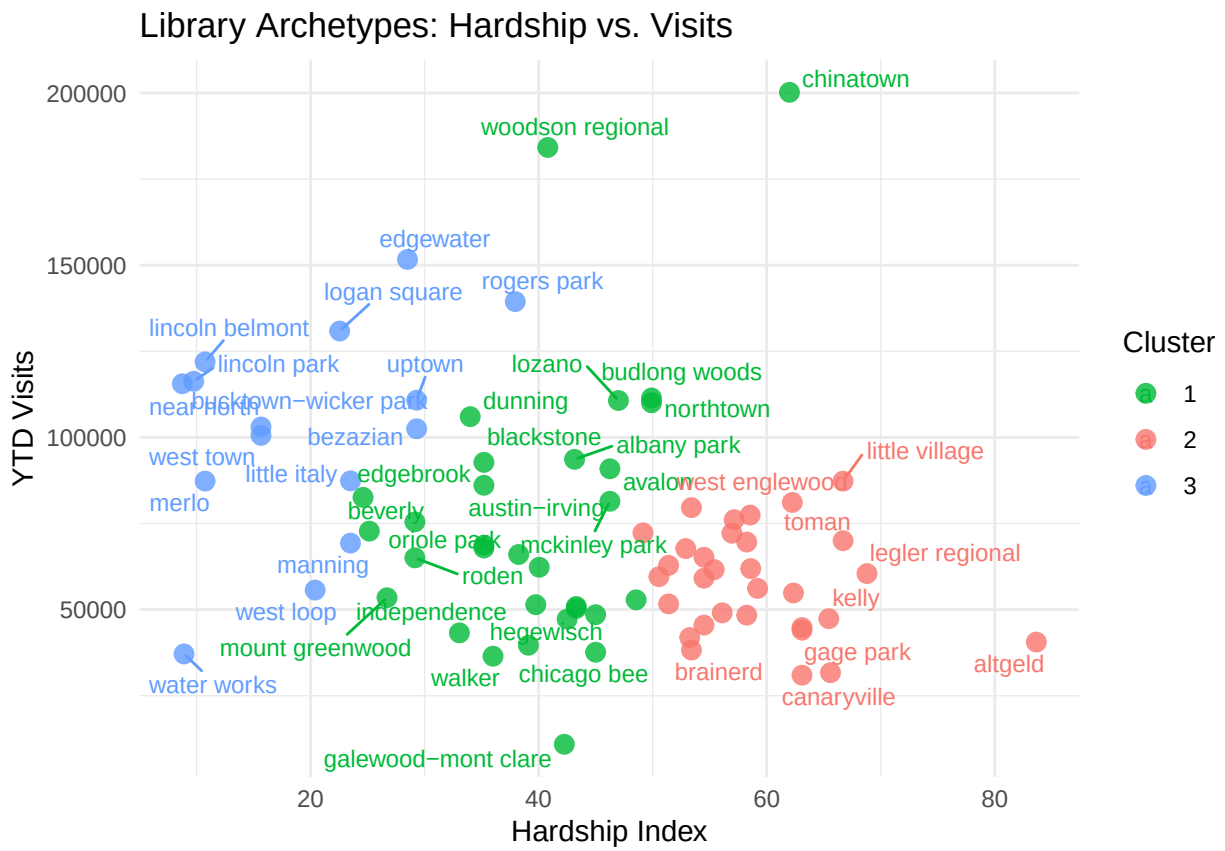
These are the clustering results and I plotted them against the average hardship index and the average yearly visits per library. It shows that there is a relationship between the amount of visits a library gets with the hardship index in that area. So ideally we would expect the branches that are in areas with high hardship to be engaging more with that community to address their needs.

```
set.seed(42)
km <- kmeans(df_scaled %>% select(-branch), centers = 3, nstart = 50)

df_clustered <- df_branch %>%
  mutate(Cluster = as.factor(km$cluster))

ggplot(df_clustered, aes(x = hardship_index, y = YTD, color = Cluster, label = branch)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c(
    "2" = "#F8766D", # Red (High Hardship)
    "1" = "#00BA38", # Green (Middle)
    "3" = "#619CFF"  # Blue (Low Hardship)
  )) +
  geom_text_repel(size = 3, max.overlaps = 10) +
  theme_minimal() +
  labs(title = "Library Archetypes: Hardship vs. Visits", x = "Hardship Index",
       y = "YTD Visits")
```

```
## Warning: ggrepel: 32 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



PART V. Analysis & Results

Feature Importance

```
print(rf_model)
```

```
## Ranger result
##
## Call:
##  ranger(formula = monthly_visits ~ ., data = final_engineered,      num.trees = 500, mtry = 4, impor
##
## Type:                             Regression
## Number of trees:                  500
## Sample size:                      9451
## Number of independent variables:  18
## Mtry:                             4
## Target node size:                 5
## Variable importance mode:         permutation
## Splitrule:                        variance
## OOB prediction error (MSE):       5356861
## R squared (OOB):                  0.799311
```

```r
importance_scores <- rf_model$variable.importance

importance_df <- data.frame(
  Feature = names(importance_scores),
  Importance = importance_scores) %>%
  arrange(desc(Importance))

print(importance_df)
```

```
##                           Feature Importance
## Pct_Working_Age   Pct_Working_Age   26094677
## Pop_Density           Pop_Density   22191358
## Pct_Black               Pct_Black   19724456
## year                         year   18586145
## percapita_income percapita_income   17890532
## Pct_Youth_0_17     Pct_Youth_0_17   17861603
## Pct_White               Pct_White   17229859
## Pct_High_Income   Pct_High_Income   16656194
## Pct_Asian               Pct_Asian   16627963
## hardship_index     hardship_index   16201424
## Pct_Senior_65       Pct_Senior_65   15113201
## area                         area   15070571
## community               community   14543626
## Pct_Hispanic         Pct_Hispanic   14075015
## branch                     branch   13870095
## Pct_Low_Income     Pct_Low_Income   13806144
## month                       month   12618212
## zipcode                   zipcode   12085445
```

Now I want to plot those features by order of importance using a lollipop graph. Each feature needed to be converted to a factor.

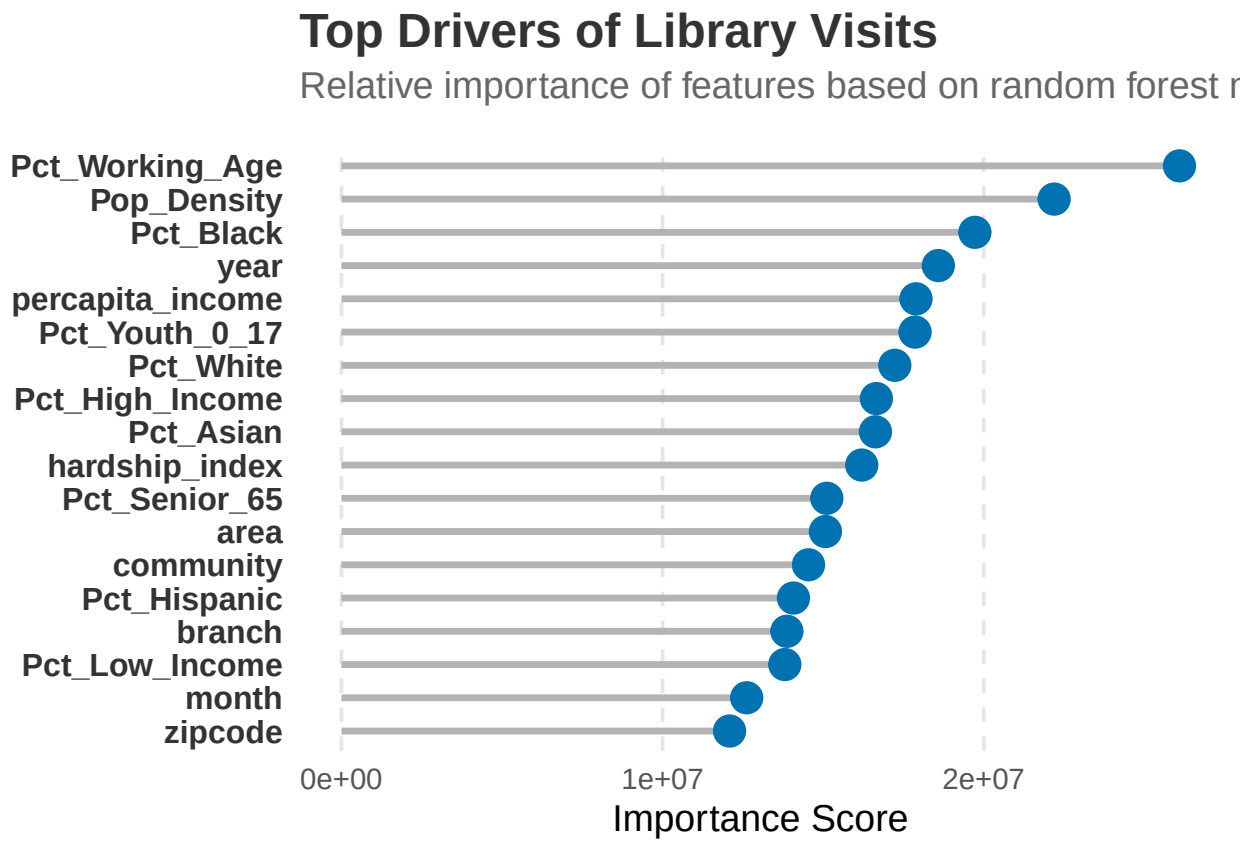
```r
importance_df <- data.frame(
  Feature = names(importance_scores),
  Importance = importance_scores
) %>%
  arrange(desc(Importance))

importance_df$Feature <- factor(importance_df$Feature,
                                levels = rev(importance_df$Feature))

ggplot(importance_df, aes(x = Feature, y = Importance)) +
  geom_segment(aes(x = Feature, xend = Feature, y = 0, yend = Importance),
               color = "gray70", size = 1.2) +
  geom_point(color = "#0072B2", size = 5) +
  coord_flip() +
  theme_minimal(base_size = 14) +
  labs(
    title = "Top Drivers of Library Visits",
    subtitle = "Relative importance of features based on random forest model",
    x = NULL,
    y = "Importance Score") +
  theme(
```

```
    plot.title = element_text(face = "bold", size = 18, color = "#333333",
                              margin = margin(b = 5)),
    plot.subtitle = element_text(size = 14, color = "#666666",
                                 margin = margin(b = 15)),
    axis.text.y = element_text(size = 12, face = "bold", color = "#333333"),
    axis.text.x = element_text(size = 11, color = "#555555"),
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_line(color = "gray90", linetype = "dashed")
  )
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



Cluster Profiles

For this section, I wanted to analyze the specific factors that differentiated each cluster from one another, specifically by focusing on variables that dealt with vulnerability, such as hardship, the senior population, and the youth population.

```r
cluster_profiles <- df_clustered %>%
  group_by(Cluster) %>%
  summarise(
    Avg_Visits = mean(YTD),
    Avg_Hardship = mean(hardship_index),
    Avg_Pop_Density = mean(Pop_Density),
    Avg_Pct_Youth = mean(Pct_Youth),
    Avg_Pct_Senior = mean(Pct_Senior),
    Branch_Count = n()
  )

print(cluster_profiles)
```

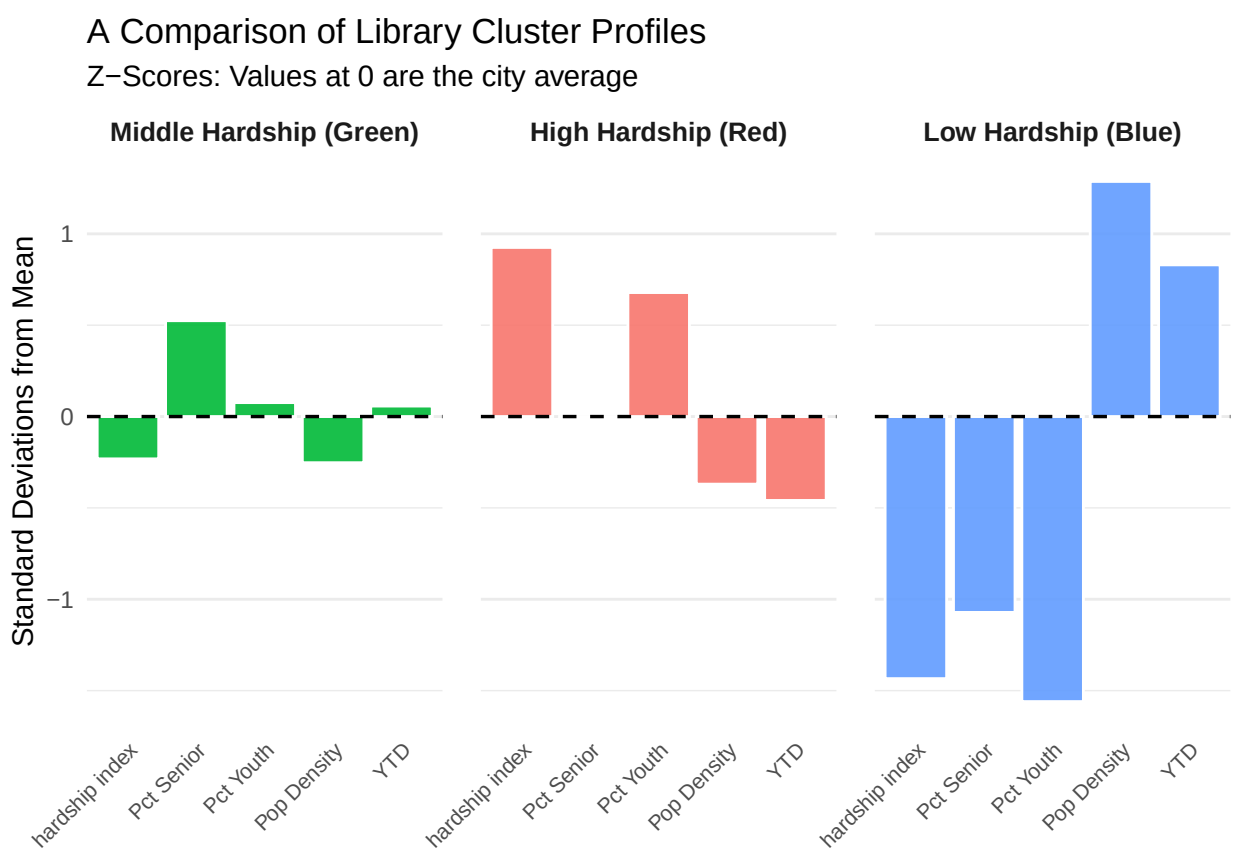
```
## # A tibble: 3 x 7
##   Cluster Avg_Visits Avg_Hardship Avg_Pop_Density Avg_Pct_Youth Avg_Pct_Senior
##   <fct>        <dbl>        <dbl>           <dbl>         <dbl>          <dbl>
## 1 1           75827.         39.7        0.000427         0.216         0.162
## 2 2           58353.         59.0        0.000398         0.248         0.140
## 3 3          101962.         19.7        0.000808         0.128         0.0970
## # i 1 more variable: Branch_Count <int>
```

To accurately compare between all these values with the city average values, the values were standardized and a faceted plot was created. This chart shows how the three clusters compare to the city average which is the dotted line in the middle. The green cluster has close to average values, but it does seem to have higher population of seniors. The red cluster shows a high hardship index, a higher percentage of youth but library visits below average. Then the blue cluster is population dense with very high visits which makes sense. There's probably more people living within walking distance to the libraries. But there's significantly less youth and senior residents.

```r
cluster_scaled_profiles <- df_clustered %>%
  select(Cluster, YTD, hardship_index, Pop_Density, Pct_Youth, Pct_Senior) %>%
  mutate(across(where(is.numeric), ~ as.vector(scale(.x)))) %>%
  group_by(Cluster) %>%
  summarise(across(everything(), mean)) %>%
  pivot_longer(cols = -Cluster, names_to = "Metric", values_to = "Z_Score") %>%
  mutate(Metric = str_replace_all(Metric, "Pct ", "% "),
         Metric = str_replace_all(Metric, "_", " "))

ggplot(cluster_scaled_profiles, aes(x = Metric, y = Z_Score, fill = Cluster)) +
  geom_col(alpha = 0.9, color = "white") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black", size = 0.6) +
  facet_wrap(~ Cluster, ncol = 3, labeller = as_labeller(c(
    "2" = "High Hardship (Red)",
    "1" = "Middle Hardship (Green)",
    "3" = "Low Hardship (Blue)"))) +
  scale_fill_manual(values = c(
    "2" = "#F8766D",
    "1" = "#00BA38",
    "3" = "#619CFF")) +
  theme_minimal() +
  labs(title = "A Comparison of Library Cluster Profiles",
    subtitle = "Z-Scores: Values at 0 are the city average",
    y = "Standard Deviations from Mean",
```

```
    x = NULL) +
  theme(legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    strip.text = element_text(face = "bold", size = 10),
    panel.spacing = unit(1, "lines"),
    panel.grid.major.x = element_blank())
```



Finally, I wanted to visualize the differences on a spatial map. I calculated the average hardship for each community and then combined it with the boundary dataset created in the beginning of the script. These would then be layered to create the map below.

```
community_hardship <- final %>%
  st_drop_geometry() %>%
  group_by(community) %>%
  summarise(Avg_Hardship = mean(hardship_index, na.rm = TRUE)) %>%
  ungroup()

chicago_poly <- chicago.comm %>%
  left_join(community_hardship, by = "community")

mapped_clusters <- library.points %>%
  inner_join(df_clustered %>% select(branch, Cluster, YTD), by = "branch")
```

Mapping the Results

Note: The code for this map runs well in Rmarkdown. However, it doesn't display when I render it in PDF since it is an interactive map. To bypass this inconvenience, I have included a picture of the plot.

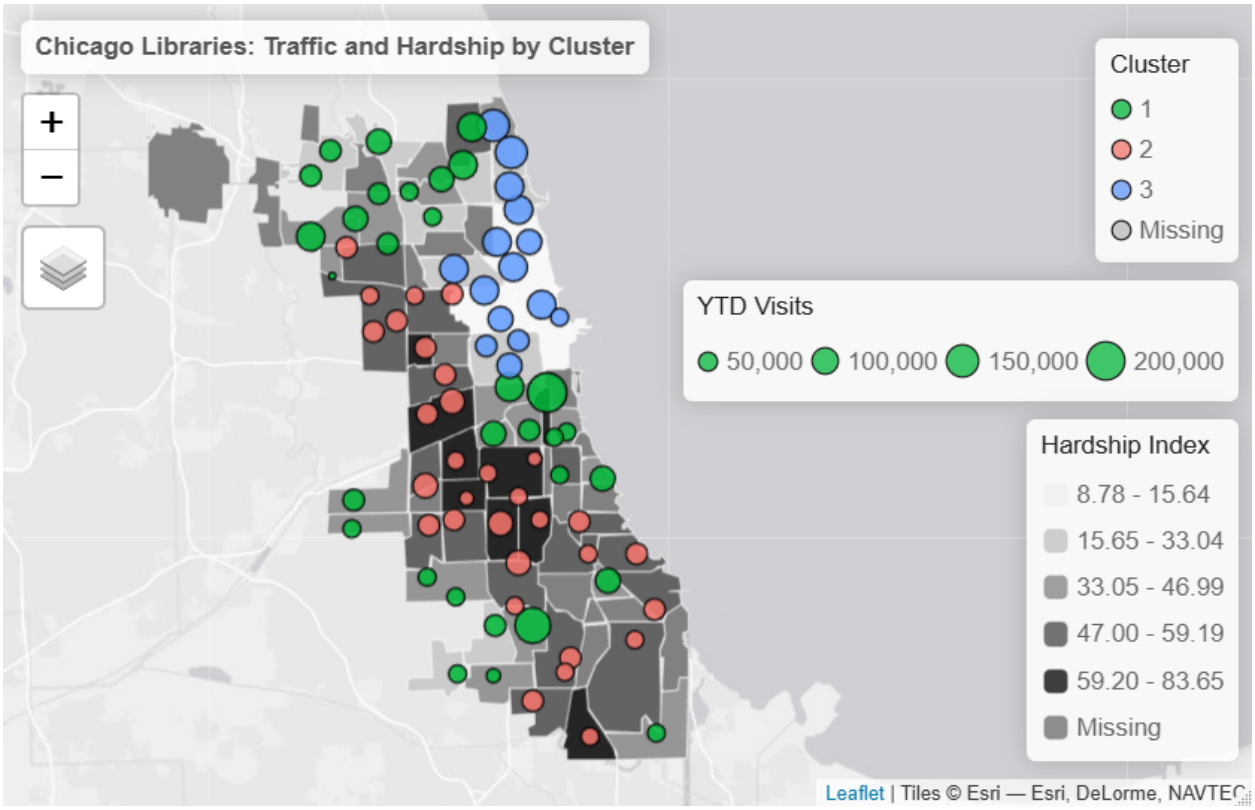


Figure 1: Chicago Libraries mapped

```
tmap_mode("view")

tm_shape(chicago_poly) +
  tm_polygons(
    col = "Avg_Hardship",
    style = "jenks",
    palette = "Greys",
    title = "Hardship Index",
    border.col = "white",
    border.alpha = 0.6,) +
tm_shape(mapped_clusters) +
  tm_symbols(
    col = "Cluster",
    size = "YTD",
    scale = 1.5,
    palette = c("#00BA38", "#F8766D", "#619CFF"),
    border.col = "black",
    border.lwd = 0.5,
    alpha = 0.85,
    title.col = "Archetype",
```

```
    title.size = "YTD Visits") +
tm_layout(
  main.title = "Chicago Libraries: Traffic and Hardship by Cluster",
  main.title.size = 1.1,
  legend.outside = TRUE,
  legend.outside.position = "right",
  legend.bg.color = "white",
  legend.bg.alpha = 0.85,
  frame = FALSE)
```

PART VI. Conclusions

In conclusion, the results of the random forest and clustering models demonstrated the most influential features that determine whether or not a library gets visited and how those variables are distributed throughout Chicago's neighborhoods. The most important variables are working age adults, population density, and a percentage of the black population.The random forest model predicted ~80% of the variance in monthly visits. Then the clustering revealed that these variables are community-dependent. Those communities in the North, which host the highest number of working age adults and is the most dense in population, receive the highest number of visits to their libraries. In contrast, the other neighborhoods to the west and south, receive less visits, and their common characteristic is that they have higher than average hardship indeces. The cluster profiles show how the green and red clusters, the middle and high hardship communities, respectively, also have a significant senior and youth population. Considering the needs of these two populations, it would be good to know how effectively they are being reached and if there are any specific barriers in place from doing so. These populations stand to benefit the most from the libraries' free services, such as educational help, community initiatives, and public health assistance.